🚦 Traffic Signal Optimization using GA and GOA

(Detailed Report – Easy Explanation)

📌 1. Introduction

Traffic congestion is a major problem in cities. Vehicles spend a lot of time waiting at traffic signals, which increases delay and fuel consumption.

This project focuses on **optimizing traffic signal timings** using two AI-based algorithms:

- Genetic Algorithm (GA)
- Grasshopper Optimization Algorithm (GOA)

The goal is to **reduce total traffic delay** by finding better signal timings.

🎯 2. Objective

The main objective of this project is:

- To **minimize traffic delay**
- To **optimize signal timings**
- To compare different approaches:
    - Normal signal timing
    - Random timing
    - Webster method (traditional)
    - GA optimization
    - GOA optimization

👉 In simple words:
We want to find the **best traffic light timings** so vehicles wait less.

🚙 3. Problem Understanding

Normally, traffic signals:

- Have fixed timings (e.g., 30 seconds each)
- Do not change based on traffic

👉 This causes:

- Long queues
- Unnecessary waiting
- Inefficient traffic flow

💡 **Our Idea**

We simulate traffic and then use AI algorithms to:

- Try different signal timings
- Find the best combination
- Reduce total delay

🧠 **4. Methodology (How the Project Works)**

Step 1: Load Dataset

We use real traffic data:

- Traffic volume (number of vehicles)

👉 This makes the simulation realistic.

Step 2: Build Road Network

- Roads are converted into a graph
- Intersections = nodes
- Roads = edges

👉 Similar to how maps work.

Step 3: Select Route

- A path is selected (like an ambulance route)
- Signals exist at intersections along this route

👉 We focus on optimizing signals on this path.

Step 4: Traffic Simulation

We simulate how traffic behaves:

🚗 **Vehicles arrive**

- Sometimes normal traffic (5–15 vehicles)
- Sometimes heavy traffic (20–40 vehicles)

🚦 **Queue formation**

- Vehicles wait at red lights
- Queue builds up

📈 **Congestion**

- More vehicles → more delay

Step 5: Delay Calculation

Total delay is calculated using:

Total Delay =

Travel Time

+ Signal Waiting Time

+ Queue Effect

Step 6: Optimization

We apply two algorithms:

🧬 **5. Genetic Algorithm (GA)**

How it works:

- Starts with random signal timings
- Combines and mutates them
- Improves over generations

👉 Like evolution:

- Best solutions survive
- Weak ones are removed

🦗 6. Grasshopper Optimization Algorithm (GOA)

How it works:

- Inspired by grasshopper swarm behavior
- Explores search space efficiently
- Finds better solutions faster

👉 Better than GA in most cases

📊 7. Comparison Methods

We compare:

Method	Description
Normal	Fixed timing (no optimization)
Random	Random signal timings
Webster	Traditional formula-based method
GA	AI optimization
GOA	Advanced AI optimization

📈 8. Results

Example result:

Normal delay: High

Random delay: Worse

GA delay: Improved

GOA delay: Best

👉 GOA gives the **lowest delay**

📉 **Graphs Generated**

- Convergence graph (how algorithm improves)
- Bar chart (comparison of methods)
- Delay vs time graph

🧠 **9. Key Concepts Used**

- Optimization algorithms
- Graph theory
- Traffic simulation
- Queue modeling
- Artificial Intelligence

⚙️ **10. Constraints**

- Green time: 5 to 60 seconds
- Total signal cycle ≤ 120 seconds

💡 **11. Special Features**

- Real dataset used
- Dynamic traffic simulation
- Burst traffic handling
- Nonlinear congestion modeling
- Multiple optimization techniques

🚑 **12. Use Case (Ambulance Scenario)**

👉 We use ambulance routing as an example.

Why?

- Delay is critical in emergencies

- Shows importance of optimization

👉 Important:
This is just an **application**, not the main objective.

🏁 13. Conclusion

- Traffic delay can be reduced using optimization
- GOA performs better than GA
- Intelligent systems improve traffic efficiency

👉 Final takeaway:

Optimized signals → less waiting → better traffic flow

🔮 14. Future Work

- Multi-intersection optimization
- Real-time traffic control
- Integration with sensors (IoT)
- Reinforcement learning models

🎤 16. Simple Summary (for teammates)

This project finds the best traffic signal timings using AI

to reduce waiting time and improve traffic flow.

🏆 Final Note

This project combines:

- Simulation
- Optimization
- Real-world application

👉 Making it a **complete and practical AI project**